

Steven Lysonski

A Boundary Theory Investigation of the Product Manager's Role

Boundary spanning theory and role theory are used to examine the product manager's role in consumer industries. A research model depicting associations between environmental uncertainty, boundary spanning, role conflict and ambiguity, job satisfaction, tension and performance, and moderating variables is developed and tested with bivariate correlations, path analysis, and ANOVA. Implications of the results are given.

SINCE its introduction by Procter and Gamble over 50 years ago, the product management system now operates in about 85% of all consumer package goods industries (Buell 1975). Despite its popularity, researchers have not probed the behavioral and communicational aspects of this dynamic activity. For example, the system's tendency to produce managerial burnout fosters rapid turnover. Obviously, behavioral issues deserve the attention of researchers. This study applies role theory in a boundary spanning context to gain an understanding of the product manager's role. Boundary spanning refers to interdepartmental and external communication.

Several aspects of a product manager's role make it important and unique. First, it is a "boundary role" at the interface between the firm and its market environment, as well as between departments within a firm. While much of a product manager's communication is informal, others are formal and routine. As one product manager remarked, "I feel like a real pol-

itician at times—I'm always lobbying for support from other departments." Hence, a product manager must deal with diverse role expectations, not only from persons in other departments, such as finance, engineering, and production, but also from outsiders such as customers, the trade, consultants, and advertising agencies.

Second, the product manager contributes to successful profit making activities. With high environmental uncertainty, he/she must adapt the product line's marketing mix to this uncertainty. For example, the unpredictable inflation of the early 1980s impinged upon decision making. Since a product manager needs the cooperation of production, sales, and research and development, he/she engages in boundary spanning activities to establish many different role relationships. Consequently, the product's success or failure affects a number of related positions whose occupants have a vested interest in attempting to influence the product manager's behavior.

Third, a product manager faces countless conflicting or ambiguous pressures as internal-external and formal-informal interfaces make their demands. As these pressures heighten, the manager's performance, satisfaction, and feeling of control are inevitably affected. Several persons interviewed expressed tense feelings of being "caught in the middle." Although

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researchers have studied the social organizational behavioral dimensions of the product manager's role (Clewett and Stasch 1975, Gemmill and Wilemon 1972, Venkatesh and Wilemon 1976), they have not yet examined its boundary spanning implications.

Boundary Spanning Concept

Because communication and integration are essential for effective performance, the product manager's boundary activities are crucial. Leifer and Delbecq (1978) note that "persons who operate at the periphery or boundary of an organization with elements outside it are called boundary spanners" (p. 41). Figure 1 depicts the product manager's communication linkages between the external environment and the internal operating environment and between departments within the firm. The product manager is a central transmitter of information about the product from the environment into the firm and vice versa.

Role theory is a promising tool for an analysis of boundary spanning activities (Adams 1976, Aiken and Hage 1972, Kahn et al. 1964, Leifer 1975, Leifer and Delbecq 1978, Miles 1975a,b, Thompson 1967). Kahn et al. (1964) were among the first to recognize the importance of managers occupying boundary positions. Thompson (1967) refined the concept of boundaries to organizational performance, while Adams (1976) applied boundary role theory to a study of labor negotiations.

The literature on product management implies the existence of boundary spanning and notes that the product manager integrates planning for a product line

through interfacing with functional departments. Luck and Nowak (1965), for example, refer to the product manager's role as "gather[ing] ideas from internal and external sources" (p. 144), while Ames (1970) recognizes that "the product manager acts as a focal point for screening all product improvement ideas that flow from the consumer, the trade, and the field selling organization" (p. 8-24). Venkatesh and Wilemon (1976) investigated the types of interpersonal power a product manager can command in boundary relationships with internal departments. The literature affirms that a product manager needs a network of internal and external linkages that are established through boundary spanning.

Problem Statement and Model Analysis

This study examines the relationships of several key behavioral variables related to the product manager's boundary role. The working model in Figure 2 presents a causal model of relevant but unconnected strands in the literature. The selected variables were refined to reflect the unique characteristics of the product manager's role and to provide actionable implications for product managers. Although the model does not provide a complete explanation of the forces operating on the product manager, it summarizes and integrates most of what is known about the relationships among environmental uncertainty, boundary spanning, role perceptions, personal outcomes, and certain moderators.

Basically, the model suggests that high degrees of

FIGURE 1
Illustration of the Boundary Concept

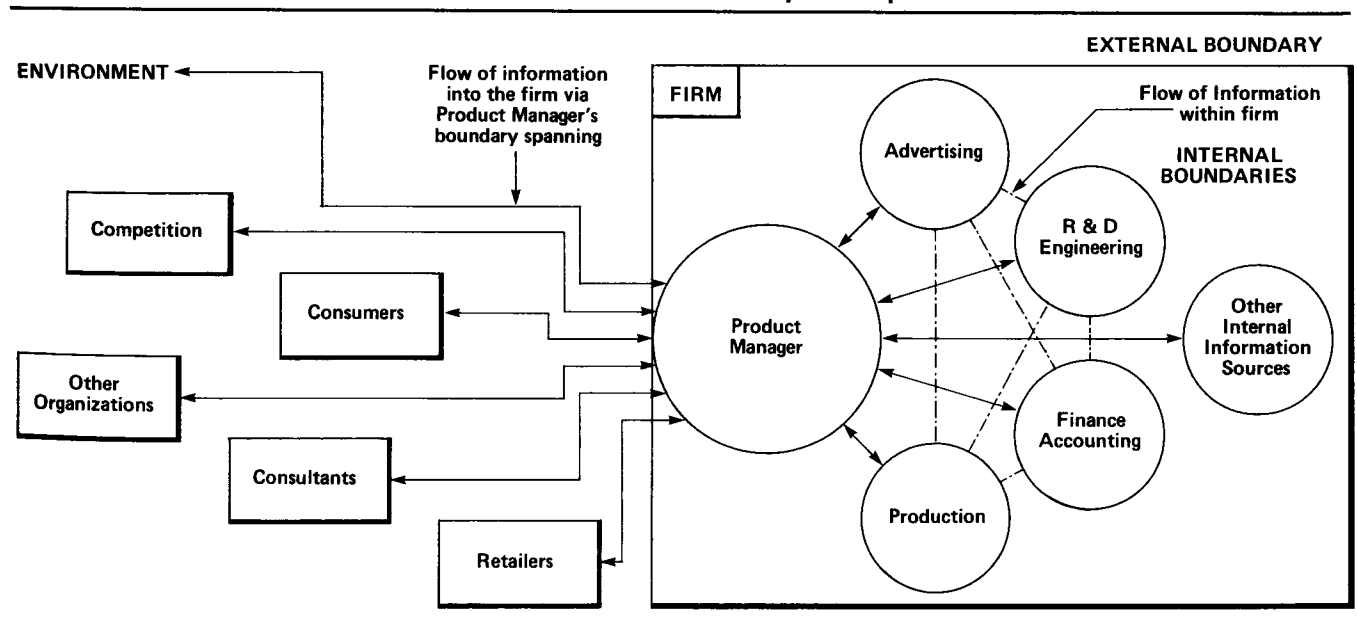
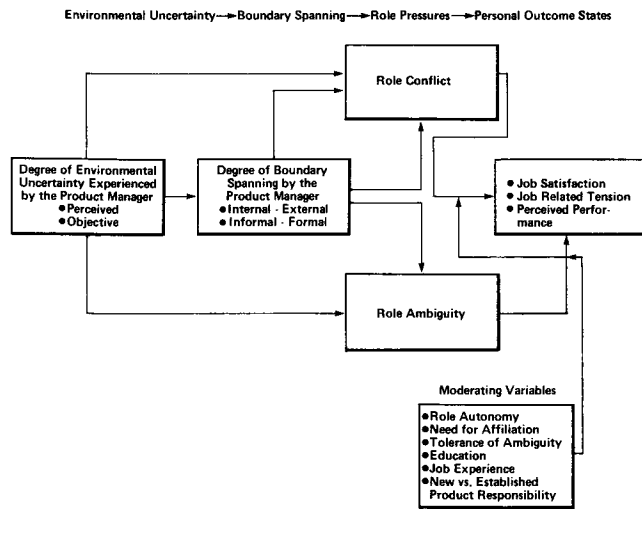


FIGURE 2
A Causal Model of the Variables in the Study



perceived and objective environmental uncertainty may lead to high levels of boundary spanning and role conflict and ambiguity for the product manager. High levels of boundary spanning in turn create role conflict and ambiguity, which lead to lower satisfaction, high tension, and perception of diminished performance for the product manager. In this model, environmental uncertainty is an exogenous variable, while the remaining variables are endogenous. Moderating variables, however, may serve to reduce the outcome states resulting from role conflict and ambiguity as portrayed in the model.

Path analysis tests the relationships in the model simultaneously, while bivariate correlations and ANOVA examine specific linkages. Finally, the managerial implications of the findings are explored.

Variables and Hypotheses

Variables in this study fall into five categories: environmental uncertainty, boundary spanning, role perceptions, personal outcome states, and moderating variables. Although the focus of this study is to explore relationships, previous research suggests a number of prior hypotheses. A discussion of each set of relationships (linkages) implied by the model concludes with the hypothesis to be tested.

Environmental Uncertainty, Boundary Spanning, and Role Perceptions

The concept of environmental uncertainty is fundamental to an understanding of the product manager's boundary spanning activities. Uncertainty involves the clarity of information, the nature of causal relationships, and the time span for feedback (Duncan 1972,

Lawrence and Lorsch 1967). Emery and Trist (1965) and Terreberry (1968) view uncertainty as turbulence, change, and the nature of interconnectedness.

According to Galbraith (1977), as uncertainty increases, so does the amount of information processing by decision makers. Thompson (1967) noted that uncertainty "appears to be [a] fundamental problem for complex organizations and coping with uncertainty as the essence of the administrative problem" (p. 159). In the same vein, Thompson suggested that boundary spanning is contingent upon the degree to which the external environment is homogeneous or heterogeneous, stable or uncertain. These ideas suggest that a product manager's response to uncertainty is to engage in more information gathering and integrating through boundary spanning (Boulton et al. 1982, Brown 1966, Conrath 1967).

In an uncertain market, rapid unpredictable changes occur, such as new technologies, increasing competitive sophistication, commodity prices, competitive entries, changes in government regulations, inflation, polarization of consumers to either high or low prices, industry oversupply, media costs, reduced margins, increased trade deal activity, and declining lead time necessary for decision making. All these uncertainties threaten the product manager's products.

Alternatively, a stable market poses fewer, more predictable threats. In general, a simpler stable market makes for easier maintenance of market share. Consequently, a product manager's ability to cope with an unpredictable market is part of his/her ability to communicate with other departments. Unlike a manager in a functional organization, a product manager can respond quickly to a rapidly changing market, thereby achieving greater effectiveness. This study points to the need for information gathering, analysis, and integration through effective boundary spanning by the product manager as a response to environmental uncertainty. A product manager uses boundary spanning to collect more information, opinions, and perspectives, thereby reducing the uncertainty or at least understanding it better. Conversely, environmental certainty creates less need for information or market monitoring. In view of these arguments, the following hypothesis is proposed:

H₁: There is a positive relationship between environmental uncertainty (perceived and objective) and boundary spanning.

The degree of uncertainty also contributes to a product manager's expectations of how other departments should cooperate (i.e., role conflict) and how he/she will be evaluated (i.e., role ambiguity). Previous research addressed role difficulties in decision making in a turbulent environment (Benson 1975, Kahn et al. 1964, Korman 1971, Roos and Starke 1981, Weed

and Mitchell 1980). Since no one has examined the linkages between environmental uncertainty and role pressures, this hypothesis posits that:

- H₂: There is a positive relationship between environmental uncertainty (perceived and objective) and (a) role conflict, and (b) role ambiguity.

Boundary Spanning and Role Perceptions of Conflict and Ambiguity

The context of the product manager's role—the activities required and the way they should be performed—is defined largely by the expectations and demands of other people. Some are fellow employees, including other product managers, his/her supervisor, advertising managers, and production personnel. Others occupy external positions such as salespeople, consultants, advertising agencies, wholesalers, and material suppliers. Consequently, the product manager encounters diverse interactions with others, both internal and external to the firm. The manager's successes or failures are apparent to others.

Role conflict is generally defined as the incompatibility of expectations associated with a role and includes several types, such as intrasender, intersender, role overload, and person role conflict (Kahn et al. 1964). Although studies have assessed the impact of boundary spanning on role conflict, none has focused specifically upon product managers. Other studies suggest that boundary spanning may have a negative impact on individuals by generating role conflict (Kahn et al. 1964, Organ 1971, Pruden and Reese 1972, Wall and Adams 1974).

Role ambiguity refers to the lack of clarity regarding what a person is required to do on a job and the subsequent performance evaluation. Research on other types of personnel has shown an association between role ambiguity and boundary spanning (Keller and Holland 1975, Keller and Szilagyi 1976).

These studies suggest the following hypotheses:

- H₃: There is a positive relationship between the product manager's boundary spanning and role conflict.
- H₄: There is a positive relationship between the product manager's boundary spanning and role ambiguity.

When role conflict occurs, the product manager experiences negative reactions due to an incompatibility of expectations and demands which cannot be simultaneously satisfied. Anxiety may arise, and the product manager may conclude that no matter how effective the performance, someone will be unhappy. As one product manager stated, "I feel like I'm in a pres-

sure cooker at times, trying to juggle all the different departments' expectations."

Kahn et al. (1964) were among the first to suggest that role conflict is a cause of job related tension, dissatisfaction, and lower levels of trust, and may affect liking and respect for role senders. Research by organizational theorists over the last 15 years tend to confirm relationships among role conflict, dissatisfaction, anxiety, and performance (Miles 1976, Miles and Perreault 1976). In particular, Miles' (1975a,b) use of crossed lagged and dynamic correlation techniques supports the theory that role conflict causes job tension and dissatisfaction.

A majority of studies on organizational behavior has been on individuals other than product managers. Several marketing studies, for example, found a negative relationship between salespersons' role conflict and both job satisfaction and performance (Bagozzi 1978; Churchill, Ford, and Walker 1974, 1976; Sheridan and Slocum 1974). Consequently, this relationship would apply to product managers, given their job pressures.

Thus the following hypotheses:

- H₅: There is an inverse relationship between role conflict and perceived performance. The greater the perceived role conflict, the lower the perceived job performance.
- H₆: There is an inverse relationship between role conflict and job satisfaction. The greater the perceived role conflict, the lower the job satisfaction.
- H₇: There is a positive relationship between role conflict and job tension. The greater the perceived role conflict, the greater the job tension.

Role Ambiguity and Personal Outcome States

A product manager suffers role ambiguity when there is insufficient information to do the job adequately, when it is unclear what peers expect, and when performance evaluation methods are ambiguous. The literature on role ambiguity in both organizational behavior and marketing suggest that this factor positively relates to feelings of tension, futility, dissatisfaction, employee turnover, and negative attitudes towards role senders (House and Rizzo 1972; Keller 1975; Miles 1976; Rizzo, House, and Lirtzman 1970). A product manager who experiences a high degree of role ambiguity, therefore, would suffer from tension, dissatisfaction, and reduced performance. In light of previous research and these arguments, the following hypotheses are proposed:

- H₈: There is an inverse relationship between role ambiguity and perceived perfor-

mance. The greater the perceived role ambiguity, the lower the perceived job performance.

H₉: There is an inverse relationship between role ambiguity and job satisfaction. The greater the perceived role ambiguity, the lower the job satisfaction.

H₁₀: There is a positive relationship between role ambiguity and job tension. The greater the perceived role ambiguity, the greater the job tension.

Moderating Variables

As illustrated in Figure 2, moderating variables may influence the relationship between role conflict, ambiguity, and personal outcome states (i.e., job satisfaction, performance, and job tension). The following moderating variables were examined: role autonomy, need for affiliation, tolerance of ambiguity, education, job experience, and new versus established product responsibility. Previous studies established the importance of these variables in moderating adverse psychological reactions (Friis and Knox 1972, Ivancevich and Donnelly 1974, Lyons 1971).

Method

The Sample

The hypotheses were tested with data obtained from questionnaires mailed to 449 product managers in consumer package goods industries, as 85% of them have product managers. Drawn from the Fortune 500, sampled industries included food, proprietary drugs, and cosmetics; soaps, cleansers, and allied products; soft drinks; tobacco; paper; and liquor—the classifications in Buell's (1975) study.

A total of 278 product managers was selected from the *Standard Directory of Advertising* and from a special 1979 issue of *Advertising Age* listing senior executives among the top 100 advertisers ("Advertising" 1979). An additional 171 subjects were obtained by contacting 293 senior executives by mail to obtain the names of product managers who would participate in the study. The total sample size, therefore, was 449.

Prior to the data collection, interviews with group product managers and product managers led to the design, selection, and modification of the study's instruments. The questionnaires were pretested on product managers and marketing managers in the Syracuse, New York, area, producing alpha coefficients in excess of .60 on all research instruments.

Questionnaires were mailed to the sample, with a cover letter explaining the research and with a promise of anonymity. To reduce evaluation apprehension and other forms of reactivity, product managers were

asked not to sign the questionnaire. A second mailing to nonrespondents produced a final return rate of 55%, or 224 completed questionnaires, of which 54 were unusable.¹ A nonrespondent check showed that neither specific industries or product categories were underrepresented. The final sample for analysis totalled 170.

Environmental Uncertainty

Two measures were used to gauge both perceived and objective uncertainty. Since some aspects of uncertainty might work through awareness and a perception process (Dill 1957/58; Downy, Hellriegel, and Slocum 1975; Huber, O'Connell, and Cummings 1975; Leblebici and Salancik 1981), the perceived measure assessed the product manager's evaluation of the environment; the objective measure, in contrast, asked product managers to state objective conditions of the environment.

There are, as yet, no definitive answers to appropriateness of a subjective versus objective measure of certainty. Anderson and Paine (1975) say: "It is generally accepted that the perceptions of environmental and internal characteristics (rather than objective characteristics of the environment) are the important properties to consider . . ." (p. 831). Snow and Darran (1975), however, argue that:

Perceptual measures are appropriate when the investigation is attempting to determine how an organization (its managers) views the behavior of the environment, because any response subsequently developed will be consistent with the perceptions. However, this approach is less appropriate when the investigation is trying to describe the nature of the environment; in these cases, it is preferable to have both perceptual and objective measures (p. 279).

The perceptual measure with 15 interval-scaled items was adapted from a scale developed by Duncan (1972) who identified the main dimensions of uncertainty in decision making. A study done by Downy, Hellriegel, and Slocum (1975) reported an alpha coefficient of .67; therefore, the Duncan uncertainty scale "appears to be methodologically adequate for basic research" (p. 626). This study's final instrument benefited from Spekman's (1979) uncertainty instrument which was based on Duncan's scale.

The objective method to quantify environmental uncertainty used six structured items: sales and price volatility of the product class, intensity of brand competition, brand switching, advertising activity, delay in feedback of product line decision after implementation, and market share fluctuations. These dimensions were identified as major causes of uncertainty for product managers.

¹Questionnaires that were incompletely answered or completed by personnel other than product managers were classified as unusable.

Boundary Spanning

Boundary spanning, defined as communication beyond a product manager's department, was measured by refining an instrument developed by Aiken and Hage (1972) and used by Leifer (1975). The final instrument evolved from the role diversity instrument developed by Snoek (1966), a boundary spanning instrument used by Keller and Holland (1975), and the dimensions relating to boundary spanning as identified by Miles (1974).

Aggregate boundary spanning was measured with the following dimensions (these components are examined separately in the analysis):

- *internal boundary spanning*: internal verbal and written communication.
- *external boundary spanning*: external verbal and written communication, e.g., customers, suppliers, trade, consultants.
- *formal boundary spanning*: formal meetings and formal written communication occurring, e.g., reports or data from others.
- *informal boundary spanning*: informal face-to-face or telephone conversations or informal written communication, e.g., informal notes, memos, and the like.
- *selected aspects*: includes travel and coordinating activities.

Product managers reported the frequencies of these boundary activities on continuous scales excluding strictly social contacts.

Role Pressures

Role conflict, defined as the degree of incongruity or incompatibility of expectations associated with a role, was measured by a 10-item Likert scale. The specific types of role conflict measured were:

- *intrasender*: incompatible expectations from a single role sender.
- *intersender*: expectations of a role sender incompatible with those of another sender.
- *person role*: incompatibility between the expectations of a product manager and the expectations otherwise associated with a role.
- *role overload*: a product manager's responsibilities exceed the time and resources available for accomplishment.

Role ambiguity, defined as inconsistencies about job responsibilities and performance evaluation, was measured with a six-item Likert scale. Both of these scales were developed and validated by Rizzo, House, and Lirtzman (1970). Several researchers reported high construct validity and other psychometric properties

of these scales which prove to be sound (Schuler, Aldag, and Brief 1977).

Personal Outcome States

Job satisfaction was measured with five Likert scale items. Product managers indicated their evaluations about promotion opportunities, chances to demonstrate individual ability and initiative, satisfactions with the position compared to other positions, whether he/she would recommend the position to others, and overall job satisfaction. This type of scale was successfully used in a number of other settings (Beehr, Walsh, and Taber 1976; Quinn and Shepard 1974).

Job tension was measured by 11 items employing Likert scales, incorporating measures of work-related worry, relief when leaving work, items from the Manifest Anxiety Scale developed by Taylor (1953), and dimensions from a study by House and Rizzo (1972).

Perceived performance was measured with two self-rating items concerning overall performance and coordinating activities for the product line. A 10-point scale was used with endpoints of "just satisfactory" and "clear excellence." Self-rating measures of performance have been found to have less leniency, less restriction of range, and less error than the purportedly more objective measures of superior ratings (Heneman 1974). Moreover, self-ratings have been found to correlate strongly with more objective measures of performance (Pym and Auld 1967). This measure also proved to be related to actual job performance of industrial salesmen (Pruden and Reese 1972).

Moderating Variables

Role autonomy is the degree to which the job provides opportunities for a product manager to exercise freedom of choice on the job. Six Likert scaled items developed by Quinn and Shepard (1974) measured this construct.

Tolerance of ambiguity is the ability to react positively to ambiguous situations. It was measured by a subset of seven Likert scaled items from Budner's instrument (1962), and found to be reliable and valid. Need for affiliation is the desire to be with others and was measured using seven Likert scaled items (Friis and Knox 1972). Job experience was measured along three experience dimensions: as a product manager, tenure with the present company, and present product responsibility.

Reliability of the Scales

This study's instruments were used successfully by a number of other researchers. It is important to note that except for objective uncertainty, these variables are essentially perceptual and, therefore, subject to a respondent's cognitive filtering process. Objective in-

TABLE 1
Scales, Items per Scale, Means, Standard Deviations, and Reliability Coefficients

Scale	Number of Items	Mean	Standard Deviation	Reliability Coefficient
Environmental uncertainty				
perceived	9	41.1	5.9	.67
objective	7	28.4	5.3	.48
Boundary spanning	11 ^a	84.4	45.3	.74
internal	4	57.7	35.2	.71
external	4	19.9	15.8	.64
informal	4	43.6	29.3	.73
formal	4	23.9	16.8	.77
others	3	15.1	11.8	.52
Role perceptions				
role conflict	10	38.6	8.5	.84
intrasender	2	7.2	2.6	
intersender	4	14.9	3.8	
person role	2	7.7	2.3	
role overload	2	8.7	2.3	
role ambiguity	6	15.6	4.7	.83
Personal outcome states				
job satisfaction	5	22.2	4.4	.75
job related tension	11	36.3	9.6	.86
perceived performance	2	14.96	2.1	.54 ^b
Moderating variables				
role autonomy	6	28.4	4.9	.80
tolerance of ambiguity	6	14.4	3.5	.60
need for affiliation	7	27.6	3.9	.52

^aThis does not add to 19 since some of the subscales overlap.

^bThis is an intercorrelation coefficient since alpha cannot be computed for two items.

struments (if they exist at all) were unavailable. The internal consistency of the variables were evaluated with Cronbach alpha coefficients, which are summaries of the internal homogeneity of a set of variables.

Table 1 presents these reliability measures and other scale information for the sample. Because reliabilities exceeding .50 are generally viewed as sufficient for research purposes (Nunnally 1967), all scales used in this study can be considered reliable.

Method of Analysis

Three forms of analysis were used. First Pearson product moment correlation coefficients tested the biserial linkages in the model and the different dimensions of boundary spanning and role conflict. This analysis is an incomplete test of the model, however, because correlations do not show each variable's relative contributions. The second, path analysis, examined the model simultaneously. To test the model fully it was necessary to reduce the number of boundary spanning and role conflict dimensions to simple constructs, since path analysis with 15 variables would be overly complicated. Similarly, the moderating variables were removed from the path analysis; instead, ANOVA examined the moderating variables.

To determine the effects of the moderating variables on the hypothesized relationships, the moder-

ating and independent variables were dichotomized into low and high groups at the midpoints. For example, the midpoint of role autonomy separated product managers into high or low role autonomy groups. The resultant high and low groups for each variable were used as the treatments for an ANOVA examination of the interaction effects. Absence of interaction indicated that the variable was not a moderator variable in the proposed relationship. Miles (1974) used a similar approach in examining moderating variables.

Linkage assessments were not based upon the criterion of statistical significance, since this is not an accepted method of determining causal significance (Duncan 1971). A path coefficient must show a magnitude of .05 or greater to be considered causally significant (Land 1969). If any path coefficient was found to be nonsignificant, the path was deleted and the system evaluated again. The basic test for the adequacy of a causal model is to compare the original correlation matrix with the reproduced correlation matrix (Kerlinger and Pedhazur 1973). In almost every case in which the reproduced correlation coefficient was compared to the original one, the discrepancy was less than .01; this suggests that the models adequately explained the relationships among the variables.

In testing the hypothesized model, which specifies direct and indirect linkages with path analysis, some standard assumptions were made. First, the relationships between the variables are assumed to be linear.

TABLE 2
Relationships between Environmental
Uncertainty and Measures of
Boundary Spanning

Measure of Boundary Spanning	Environmental Uncertainty	
	Perceived	Objective
Aggregate boundary spanning	.11 ^a	-.02
informal boundary spanning	.10 ^a	-.03
formal boundary spanning	.10 ^a	.00
internal organizational boundary spanning	.10 ^a	-.04
external organizational boundary spanning	.03	.04

^a $p \leq .10$.

Examination of scatter plots of the residuals indicated the residuals were randomly dispersed and that the relationships examined in the study were most likely linear. Moreover, the fact that the scatter plots indicated no serious problem with "outliers" suggested that the model has no misspecification problems. It must be recognized that the results are limited to the extent that some variables are omitted from the model specification in this study. Second, it is assumed that the residuals of a given endogenous variable are not correlated with other system variables nor other variables. Third, the causal system posited by the model is viewed as recursive (i.e., there is no reciprocal causation (Wright 1960)). Lastly, the variables are assumed to be measured on interval scales. To the extent that the scales are ordinal, the results may be slightly biased, although multiple regression is still the recommended procedure (Bohrnstedt and Carten 1971).

Results

Correlation Analysis

Table 2 contains the correlations between a product manager's environmental uncertainty (perceived and objective) and boundary spanning. Although perceived environmental uncertainty is weakly associated with the dimensions of boundary spanning, objective uncertainty is not correlated with boundary spanning. Hence H_1 receives support only for perceived uncertainty.

The correlations between environmental uncertainty and role pressures are presented in Table 3. The correlations are significant only for perceived uncertainty, while only objective uncertainty is related to intersender conflict. The results, therefore, support H_2 that uncertainty is associated with role conflict and ambiguity only for perceived uncertainty.

Table 4 shows that the correlations between the dimensions of role conflict and measures of boundary spanning are all statistically significant, thus uphold-

ing H_3 . It also shows that role ambiguity is not associated with boundary spanning except for internal organizational boundary spanning; hence, H_4 cannot be supported. Table 5 presents the correlations between role pressures and personal outcome states. In nearly every case, role conflict including its dimensions and role ambiguity is adversely associated with job satisfaction, job related tension, and perceived performance. Hence hypotheses H_5 to H_{10} are upheld.

Results of the Path Analysis

Figure 3 presents the path coefficients when perceived environmental uncertainty is the only exogenous variable. The results indicate that environmental uncertainty links strongly to role conflict ($p = .251$) and role ambiguity ($p = .237$), upholding H_2 , while the connection between perceived uncertainty and boundary spanning is low ($p = .114$). Hence, H_1 (perceived uncertainty leads to increased boundary spanning) is upheld, but weakly.

As hypothesized in H_3 , boundary spanning was found to be related to role conflict ($p = .181$). Since the path coefficient between boundary spanning and role ambiguity is less than .05, H_4 was rejected.

The hypotheses (H_5 to H_{10}) that role conflict and ambiguity lead to adverse psychological reaction (of tension, dissatisfaction, and reduced perceived performance) are also upheld. The connection between perceived performance and role conflict, however, has a path coefficient of $-.066$, and the path coefficient between tension and role ambiguity registered a .052 path coefficient; both are low and near the .05 cutoff suggested by Land (1969).

While most of the hypotheses were upheld, unexpected findings occurred in one of the models. In Figure 3 the strong relationship between job satisfaction and perceived environmental uncertainty required an additional path in the model. This linkage ($p = .178$) suggests that perceived environmental uncertainty is causally related to an increase in job satisfaction. In addition, a linkage was found between job satisfaction and boundary spanning ($p = .061$),

TABLE 3
Environmental Uncertainty and Role Pressures

Role Pressures	Environmental Uncertainty	
	Perceived	Objective
Role conflict	.29 ^a	.07
person-role conflict	.26 ^a	.06
intersender conflict	.25 ^a	.16 ^c
intrasender conflict	.17 ^b	-.03
role-overload conflict	.19 ^b	-.04
Role ambiguity	.23 ^a	.09

^a $p \leq 0.001$.

^b $p \leq 0.01$.

^c $p \leq .05$.

TABLE 4
Relationships between Measures of Boundary Spanning and Role Conflict and Role Ambiguity

Measures of Boundary Spanning	Role Conflict	Role Perceptions				Role Ambiguity
		Person-role conflict	Intersender conflict	Intrasender conflict	Role-overload conflict	
Aggregate boundary spanning	.28 ^a	.18 ^b	.30 ^a	.22 ^a	.11	.00
informal boundary spanning	.14 ^c	.05	.17 ^b	.12 ^c	.05	.10
formal boundary spanning	.24 ^a	.13 ^c	.22 ^a	.24 ^b	.01	.05
internal organizational boundary spanning	.22 ^a	.13 ^c	.23 ^a	.17 ^b	.10	.13 ^c
external organizational boundary spanning	.12 ^c	.00	.13 ^c	.16 ^c	.04	-.04

^ap ≤ 0.001.

^bp ≤ 0.01.

^cp ≤ 0.05.

implying that boundary spanning lowers job satisfaction.

Figure 4 presents the path coefficients for the models when objective uncertainty is used as the exogenous variable. The results of these models are almost equivalent to the models with perceived uncertainty, with one major difference; objective uncertainty is not associated with boundary spanning. Hence, H_1 is not upheld for objective uncertainty. Moreover, the linkages between objective uncertainty and role pressures of conflict and ambiguity are rather low compared to the models in Figure 3 for perceived uncertainty; nonetheless, H_2 is supported for objective uncertainty.

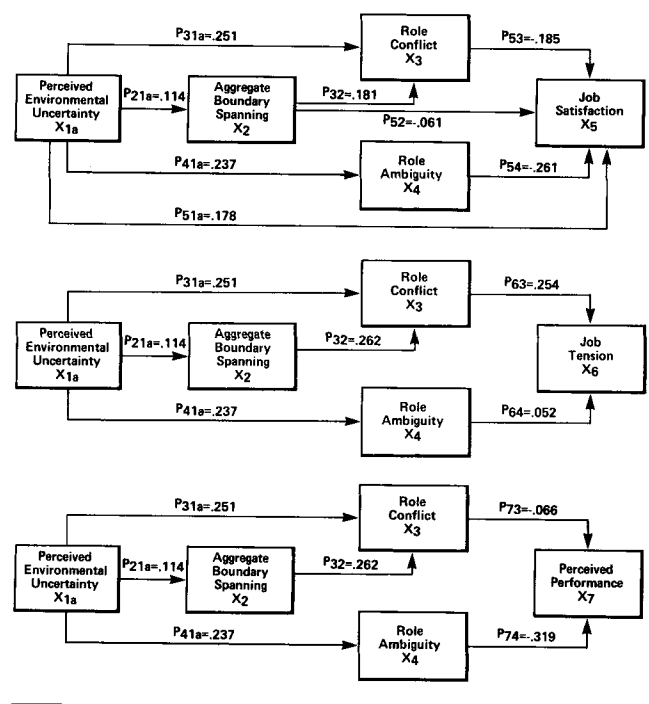
H_3 is supported given that the path coefficient between boundary spanning and role conflict is .283. H_4 was not supported, since the path coefficient between boundary spanning and role ambiguity was less than .05. Last, the hypotheses concerning role pressures and personal outcome states (H_5 to H_{10}) are accepted, since the paths between both role conflict and ambiguity and tension, satisfaction, and perceived performance are above .05. As in the perceived uncertainty models, the paths between role ambiguity and tension ($p = .052$) and between role conflict and perceived performance ($p = -.066$) are rather low, implying a

weak connection. Just as in the analysis for perceived uncertainty, boundary spanning was negatively associated with job satisfaction ($p = -.051$). Unexpectedly, objective environmental uncertainty was positively linked to job tension ($p = .098$).

Moderating Variables: ANOVA

The findings on the moderating variables in Table 6 show mixed results. Role autonomy was found to

FIGURE 3
Perceived Environmental Uncertainty Models



STRUCTURAL EQUATIONS

$$\begin{aligned}
 X_2 &= B_{21a}X_{1a} \\
 X_3 &= B_{32}X_2 + B_{31a}X_{1a} \\
 X_4 &= B_{41a}X_{1a} \\
 X_5 &= B_{53}X_3 + B_{54}X_4 + B_{52}X_2 + B_{51a}X_{1a} \\
 X_6 &= B_{63}X_3 + B_{64}X_4 \\
 X_7 &= B_{73}X_3 + B_{74}X_4
 \end{aligned}$$

TABLE 5
Relationships between Role Conflict and Role Ambiguity and Job Satisfaction

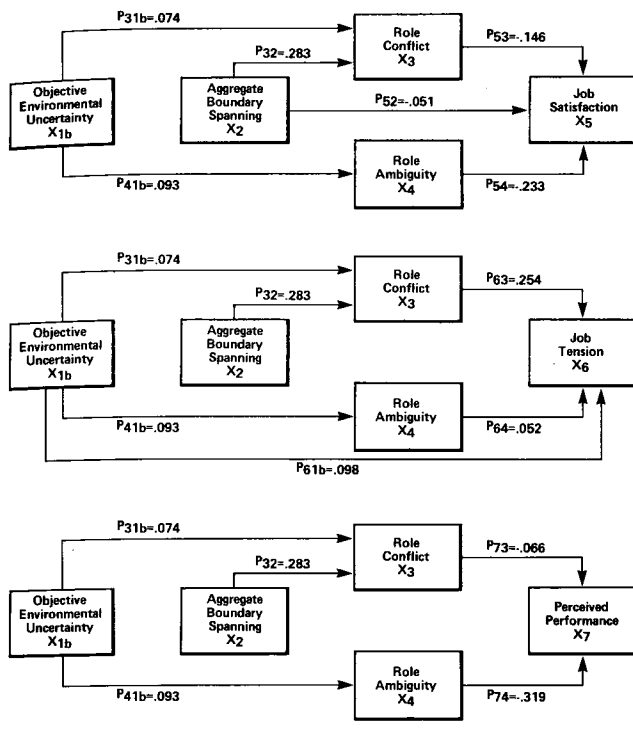
Role Perceptions	Job Satisfaction	Job Related Tension	Perceived Performance
Role conflict	-.24 ^a	.28 ^a	-.18 ^b
person-role conflict	-.25 ^a	.18 ^b	-.08
intersender conflict	-.20 ^a	.24 ^a	-.17 ^c
intrasender conflict	-.24 ^a	.24 ^a	-.17 ^c
role overload conflict	-.01	.17 ^c	-.08
Role ambiguity	-.28 ^a	.15 ^c	-.34 ^a

^ap < 0.001.

^bp < 0.01.

^cp < 0.05.

FIGURE 4
Objective Environmental Uncertainty Models



STRUCTURAL EQUATIONS

$$\begin{aligned}
 X_3 &= B_{32}X_2 + B_{31b}X_{1b} \\
 X_4 &= B_{41b}X_{1b} \\
 X_5 &= B_{53}X_3 + B_{54}X_4 + B_{52}X_2 \\
 X_6 &= B_{63}X_3 + B_{64}X_4 + B_{61b}X_{1b} \\
 X_7 &= B_{73}X_3 + B_{74}X_4
 \end{aligned}$$

moderate only the role ambiguity–satisfaction relationship. Since role autonomy gives the product manager latitude in decision making, it seems that this autonomy may maintain his/her satisfaction when role ambiguity is present.

Need for affiliation had a definite moderating influence; it lowered the role conflict–perceived performance relationship, increased the role ambiguity–job related tension relationship, and lowered the role ambiguity–perceived performance relationship. Since need for affiliation implies that the product manager prefers to be with others, the manager may expect some role conflict in dealing with them. However, a higher need for affiliation may provide greater tolerance for this conflict, while increasing job tension from role ambiguity. Tolerance of ambiguity had only one moderating effect on the role conflict–job related tension relationship by reducing tension in this relationship.

Education was effective in reducing the role conflict–job satisfaction relationship and role ambiguity–job tension relationship. Education may give the product manager greater confidence and expertise in dealing with role conflict and ambiguity. Consequently, these

role perceptions have less impact upon educated product managers.

Experience was shown to have a moderating influence on some of the relationships. For example, experience as a product manager increased the role conflict–job tension relationship and lowered the role ambiguity–perceived performance relationship. Experience with the present company as a product manager had no moderating influence on the relationships examined. However, experience with the present product as a product manager increased the role conflict–job tension and the role ambiguity–job tension relationships. Therefore, certain experiences seem to heighten the adverse reactions of the product manager to role conflict. Experience implies that the product manager may have developed more personal relationships with others and as a result, may react more personally to the conflicting expectations of others and may experience more negative personal outcomes. New versus established product responsibility had no moderating influence.

Limitations

Due to the cross-sectional data, the time sequence of the relationships among the variables could not be clearly assessed. The use of product managers from many companies at a single time period may have led to sampling artifacts and to the exclusion of temporal dynamics of the system. The results, therefore, should not be viewed as proof of the causal model but rather as evidence supporting many of the relationships in the a priori causal scheme. A longitudinal design is necessary to test temporal suppositions, although a cross-sectional investigation serves as an exploratory step for establishing face validity of hypotheses to warrant future longitudinal studies.

The measures were obtained at the same time by similar scaling techniques. Method variance, therefore, may have inflated some relationships. Rather than using reactive instruments, more objective measures of conflict, ambiguity, and performance would add an important dimension to this study and to such research. Such an improvement would provide a better basis for management prescriptions and from a methodological perspective, would reduce potential bias from method variance.

Discussion

Despite these limitations, the study has significant managerial and research implications. The four key contribution areas that emerge are:

- complexity of environmental uncertainty
- possible deficiencies in the product management system

TABLE 6
Summary Table of the ANOVA Tests on the Moderating Variables

	F Statistic for Interaction Effect (df = 1/166)
Role Conflict–Satisfaction Relationship	
Role autonomy	.144
Need for affiliation	.63
Tolerance of ambiguity	.004
Education	6.04 ^a (increased satisfaction)
Experience as a product manager	.25
Experience with present company as a product manager	.72
Experience with present product as a product manager	.15
New versus established product responsibility	.08
Role Conflict–Job Related Tension Relationship	
Role autonomy	.92
Need for affiliation	.50
Tolerance of ambiguity	4.74 ^b (decreased tension)
Education	.18
Experience as a product manager	7.42 ^c (increased tension)
Experience with present company as a product manager	.72
Experience with present product as a product manager	6.50 ^a (increased tension)
New versus established product responsibility	.39
Role Conflict–Perceived Performance Relationship	
Role autonomy	.06
Need for affiliation	7.02 ^c (increased performance)
Tolerance of ambiguity	.41
Education	.05
Experience as a product manager	.22
Experience with present company as a product manager	1.01
Experience with present product as a product manager	.37
New versus established product responsibility	.04
Role Ambiguity–Satisfaction Relationship	
Role autonomy	9.38 ^c (increased satisfaction)
Need for affiliation	.27
Tolerance of ambiguity	.13
Education	.24
Experience as a product manager	.42
Experience with present company as a product manager	1.06
Experience with present product as a product manager	.04
New versus established product responsibility	.32
Role Ambiguity–Job Related Tension Relationship	
Role autonomy	.28
Need for affiliation	5.83 ^a (increased tension)
Tolerance of ambiguity	2.60
Education	7.25 ^c (lowered tension)
Experience as a product manager	.05
Experience with present company as a product manager	.41
Experience with present product as a product manager	6.20 ^c (increased tension)
New versus established product responsibility	.89
Role Ambiguity–Perceived Performance Relationship	
Role autonomy	.31
Need for affiliation	3.83 ^b (increased performance)
Tolerance of ambiguity	.02
Education	.03
Experience as a product manager	4.85 ^b (increased performance)
Experience with present company as a product manager	.15
Experience with present product as a product manager	.34
New versus established product responsibility	.26

^ap ≤ 0.01.

^bp ≤ .05.

^cp ≤ .001.

- harmful effects of role conflict and ambiguity
- the value of path analysis

The results show that perceived environmental uncertainty is weakly related to a product manager's boundary spanning, while objective uncertainty was not related. There are several possible explanations. For example, one product manager reported that communication activities depended on the type of uncertainty and whether it was new or a customary aspect of the job. A group product manager suggested that uncertainty is not a major impulse for increased communication. Instead, a product manager has a given propensity to communicate (i.e., boundary spanning) regardless of environmental uncertainty. Some product managers communicate more than others, even in a static environment. One group product manager stated that "if a product manager spends more than half of the day in his office, he's not collecting enough information about his product." This manager encourages his product managers to interact as much as possible with external (e.g., the trade, customers) and internal (e.g. finance, production, R&D) sources.

Secondly, a weak association between environmental uncertainty and boundary spanning may signal deficiencies in the product management system. In particular, research by Hise and Kelly (1978) strengthens the observation that product managers lack sufficient external sources of information. In effect they argue that product managers are not engaging in adequate levels of external boundary spanning. Research by Tushman (1979) applicable to product management suggests that high performing project teams relied more on boundary spanning individuals to attend to informational needs when environmental uncertainty was high. Aldrich and Herker (1977) suggest that "an organization's ability to adapt to environmental contingencies depends in part on the expertise of the boundary role incumbent in selecting, transmitting, and interpreting information originating in the environment" (p. 229).

Third, the results underscore the notion that perceived environmental uncertainty is responsible for role conflict and ambiguity as boundary spanning. Much has been written on the effects of these role pressures on an individual's sense of well-being. These role pressures were found to be associated with adverse personal outcomes such as lower perceived performance, lower job satisfaction, and increased tension. These findings confirm research that found similar associations for other types of personnel (Van Sell, Brief, and Schuler 1981).

Finally, these results demonstrate the value of path analysis in studying marketing systems. Correlation is

a useful point of departure but does not reveal how the variables are simultaneously related. Path analysis provides a view of the variables as a system of relationships.

Implications

Behavioral Consequences of Role Pressures

In some cases, tensions from role conflict and ambiguity may not adversely affect a product manager's performance. In fact, moderate tension may be an energizer resulting in increased motivation (Kahn et al. 1964). High levels of stress, however, may produce effects detrimental to both the product manager and the firm. Kahn et al. (1964) also say that extreme forms of role conflict and ambiguity affect physical and psychological well-being. This suggests that these forces should be kept at manageable levels. Since the product manager interacts with others, role conflict is inevitable. Therefore, the issue is not to eliminate role conflict and ambiguity but to reduce them to manageable levels (Ivancevich and Matteson 1980, Kets de Vries 1979).

Methods to Mitigate Role Pressures

Several group product managers suggested that better understanding of the company's objectives would help allay role conflict and ambiguity. It is also important that the vice president of marketing communicate company objectives to product managers. Some product managers believe they must maximize a brand's sales, even though company objectives do not always state this. Consequently a product manager may be driven by personal goals to be promoted so that "a product manager's ego might get in the way of reality," according to one group product manager. Oftentimes novice product managers expect their jobs to be challenging and exciting, yet this is the exception, since the role involves many programmed activities.

Kahn et al. (1964) suggest methods to reduce role conflict and ambiguity including (1) changes in the firm's organizational structure, (2) development of new criteria for selection and placement of personnel, (3) increasing the individual's tolerance and coping abilities, and (4) strengthening the interpersonal bonds among organizational members. The burden for using these and other methods to diminish the negative effects of role conflict and ambiguity rests on a company's senior management. The present study has shown that different types of role conflict (i.e., person role conflict, intersender conflict, intrasender conflict, and role overload conflict) are associated with adverse personal outcome states. Various methods must be used

to manage the different types of conflict faced by product managers. For example, if a product manager experiences intersender conflict, more and better communication with senior management seems necessary. In written or oral communication, product managers should be informed of changes in company policies and, importantly, why these changes are being made.

The setting of brand objectives also helps lessen role conflict and ambiguity. Clearcut objectives for product managers are crucial. Senior managers reported that these goals should be set mutually, not by fiat. Such a dialogue helps clarify a product manager's role.

A formal review process for measuring a product manager's performance in meeting these objectives may also reduce role ambiguity. For example, one group product manager suggested that a product manager's performance be reviewed every six months. Although these reviews need not be recorded, they serve as "pragmatic reviews in and of themselves." At a minimum, frequent discussions between group product managers and product managers would provide freer exchange of differing views.

Senior management could reduce role conflict resulting from interdepartmental relations. If senior management is sensitive to potential conflict between managers and other departments, they can clarify the role a product manager is to perform in decision making. This would reduce the likelihood of intersender conflict for a product manager. Another way to lower this form of role conflict is a formal process of conflict resolution to maintain understanding between the boundary interests of product managers and other departments.

Senior management can mitigate potential role conflict and ambiguity during recruitment and training.

The challenge is not only to minimize role conflict and ambiguity inherent in the product management system but to identify those product managers most capable of dealing with these difficulties. Analysis of the moderating variables, for example, showed that a high affiliation need is beneficial. Product management training could expose novices to simulated pressures through role playing and case studies.

Summary

This study not only reinforces existing knowledge about individuals who span organizational boundaries but provides a different approach in examining the product management system. Many more issues, however, need to be studied. For example, it would be useful to know how instrumental boundary spanning is in the innovation process. Because the product manager aids in the dissemination of information, he/she may be facilitating transfer of new ideas and technology across boundaries, thus stimulating the innovation process. It is also possible that the product manager functions as a gatekeeper of information in conducting boundary spanning. Other avenues for research include the means by which the product manager can monitor the environment, the information processing mechanisms used, and how the successful product manager copes with stress and conflict.

Since the product management system has scored significant gains in administering complex product lines, this research illuminates several behavioral aspects of the system. Environmental change and turbulence are likely to accelerate in the future; thus, additional studies that examine the adaptation of various product line administration systems to changing environmental forces will be needed. This study is a step in that direction.

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